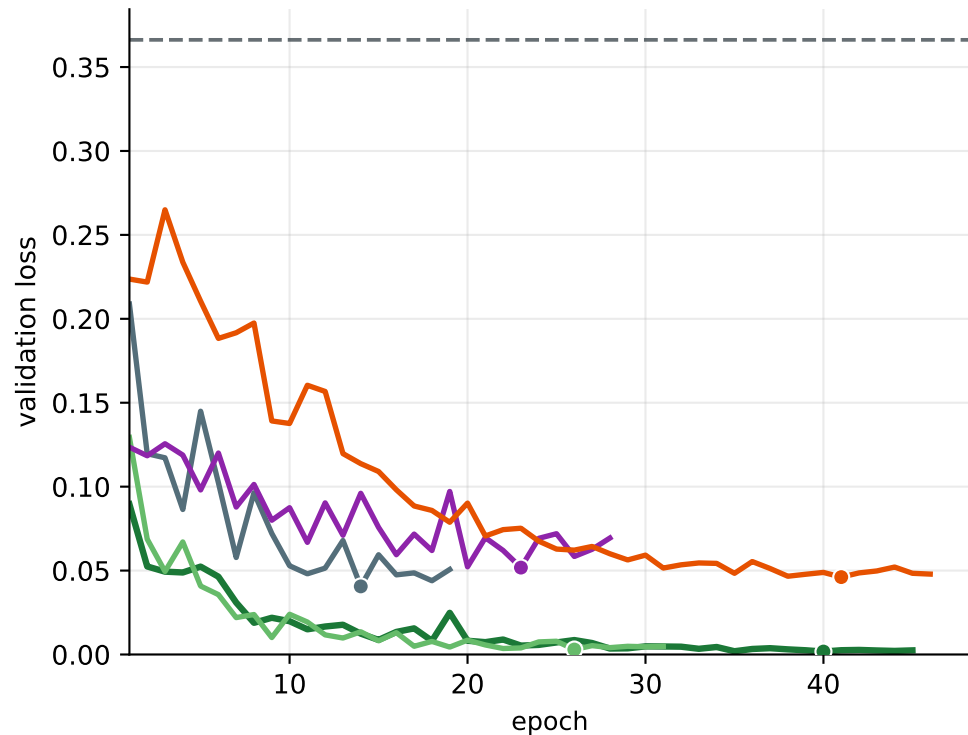
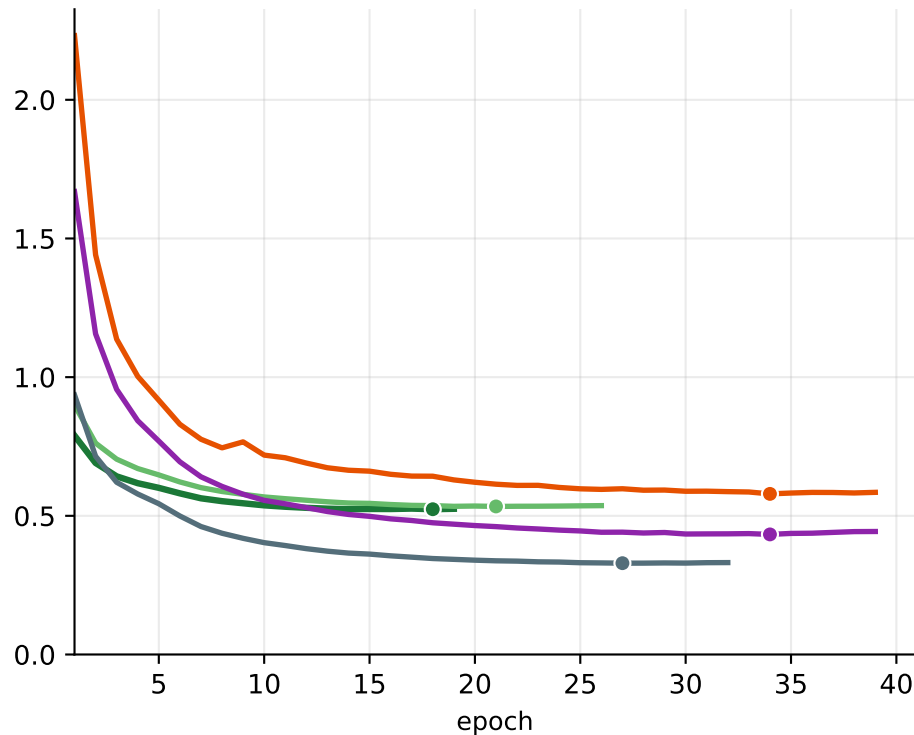


Validation loss across all tasks — identical 50-epoch protocol (patience-5)

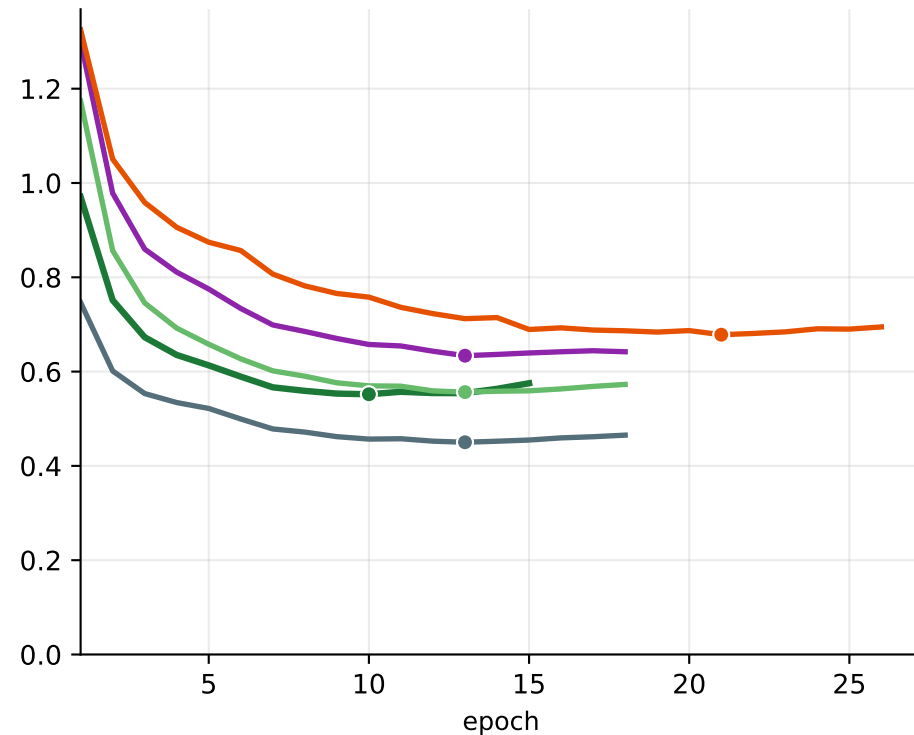
Stage-1 TSQA



Stage-3 HAR-CoT



Stage-4 Sleep-CoT



Mamba-1.4b Mamba-370m SP Llama+bins Flamingo --- random = 0.37

Dots mark each model's best epoch; TSQA dashed line = random 3-way chance. On TSQA the SSM+token models converge far lowest. For the CoT tasks (HAR/Sleep), validation loss is teacher-forced cross-entropy over the full reasoning text and DIVERGES from answer macro-F1 (Table 1): a model can match the reference reasoning closely yet pick the wrong final label, so SP's lower CoT text-loss does not translate to higher F1— the Mamba models still lead the answer metric.